FIITJEE

ALL INDIA TEST SERIES

FULL TEST – X

JEE (Main)-2024

TEST DATE: 29-03-2024

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. The two plates of a parallel plate capacitor with no dielectric are connected to a battery with constant emf. Now a dielectric of dielectric constant K is inserted to fill the entire space between the plates without removing voltage source. Find the **INCORRECT** options.
(A) The energy stored in the capacitor will become k times.
(B) The electric field between plate will become 1/k times.
(C) The force of attraction between plates will become k^2 times.
(D) Charge on capacitor will become k times.

2. Match List I and List II

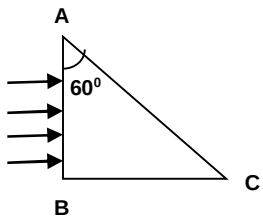
	List – I		List-II
(a)	Ultraviolet rays	(1)	Radar system
(b)	Infrared rays	(2)	Green house effect
(c)	X-rays	(3)	Study crystal structure
(d)	Microwaves	(4)	Sterilizing surgical instrument

(A) a – 2, b – 1, c – 4, d -3 (B) a – 4, b – 3, c – 2, d -1
(C) a – 4, b – 2, c – 3, d -1 (D) a – 2, b – 4, c – 1, d -3

3. A non iso tropic solid metal cube has coefficients of linear expansion given by 40×10^{-6}/°C along X-axis, 3×10^{-6}/°C along Y-axis and 7×10^{-6}/°C along Z-axis. What is coefficient of volume expansion of cube?
(A) 5×10^{-5} /°C (B) 94×10^{-17} /°C
(C) 4×10^{-5} /°C (D) 21×10^{-12} /°C

4. Magnetic materials used for making magnets in a transformer (T) and permanent magnets (P) have different properties. Out of the following which property best matches for the type of magnet required
(A) T : large retentivity large coercivity (B) T: large retentivity small coercivity
(C) P: large retentivity large coercivity (D) P: small retentivity large coercivity

5. Parallel light incident on prism which contains (Blue, green and red colour. Refractive index of prism for them are 1.3, 1.2 and 1.1 respectively. Which colour will come out from face AC.
(A) Red (B) Red and Black
(C) Red and green (D) all colour



6. A small current element of length 'dl' carrying current in positive x-axis is placed at [0, 2, 2] m. if magnetic field at origin is $\vec{B}_1$ and at position (0, 4, 4) is $\vec{B}_2$ then
(A) $|\vec{B}_1| > |\vec{B}_2|$ (B) $\vec{B}_1 = \vec{B}_2$
(C) $\vec{B}_1 + \vec{B}_2 = \vec{0}$ (D) $\vec{B}_1 . \vec{B}_2 = 0$

7. Time period of spring block system for diagram -A is T_1, for diagram -B it is T_2. Find out time period of spring block system for diagram C.

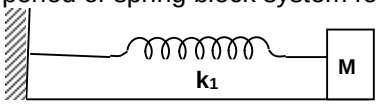


Diagram - A

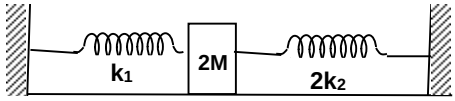


Diagram - B

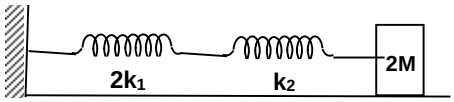
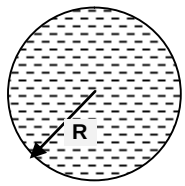


Diagram - C

(A) $\sqrt{T_1^2 + \frac{2T_1^2T_2^2}{T_1^2 - 2T_2^2}}$ (B) $\sqrt{T_1^2 + T_2^2}$

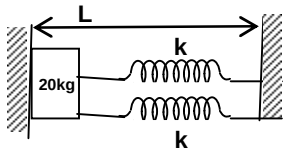
(C) $\sqrt{T_1^2 + \frac{4T_1^2T_2^2}{2T_1^2 - T_2^2}}$ (D) $\sqrt{T_2^2 + 4\frac{T_1^2T_2^2}{T_1^2 + T_2^2}}$

8. A sphere is filled with liquid. Radius of sphere is R. This sphere is accelerated horizontally with acceleration $\sqrt{3}$ g . Then find out correct statement.



(A) Pressure difference between top and bottom of liquid is more than $2\rho gR$.
(B) On a imaginary horizontal line passing from centre have same pressure.
(C) On a imaginary line which makes angle 30^0 with horizonal pressure is same.
(D) Maximum pressure difference between two position in liquid is $4\rho gR$

9. A cubical block of 20 kg and side length 0.2 m is pressed against a vertical surface by springs of unstretched length L = 1 m, the coefficient of friction between the block and the vertical surface is 0.2. Choose correct option (k is spring constant, g = 10 m/s^2).



(A) Block will be in equilibrium if k = 3000 N/m
(B) Block will be seen equilibrium if k = 500 N/m.
(C) Block cannot remain in equilibrium
(D) friction on block is 350 N if k = 1400 N/m.

10. Block B of mass m is placed on plank A of mass 4 m coefficient of friction between A and B is μ. Coefficient of friction between plank and ground is $\mu/3$. Block B is given velocity v in horizontal direction and A is given 2v is opposite direction, find ratio of magnitudes of initial acceleration of A and initial acceleration of B.



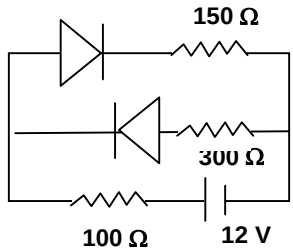
(A) 7/8 (B) 2/3
(C) 5/3 (D) 1/3

11. Force (F) on an object depends on kinetic energy k and x-coordinate of particle which is given by $F = \alpha x + \frac{\alpha}{\sqrt{\beta}}k^2$.

α and β are positive constant find dimension of β.

(A) $[ML^{-1}T^{-3}]$ (B) $[M^3L^{-4}T^{-2}]$
(C) $[M^4L^6T^{-8}]$ (D) $[M^{-18}T^{-2}]$

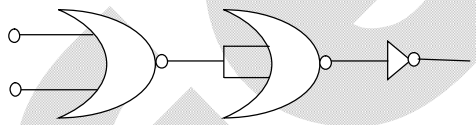
12. What is current in circuit.



(A) 60 mA (B) 30 mA
(C) 15 mA (D) 48 mA

13. A ball is dropped on horizontal floor from a height 128 m. Coefficient of restitution between ball and ground is $(1/2)^{1/8}$. Then height attained by ball
(A) after two collision is 64 m (B) after four collision is 64 m
(C) after two collision is 16 m (D) after 4 collision is 32 m.

14. Given network is equivalent to

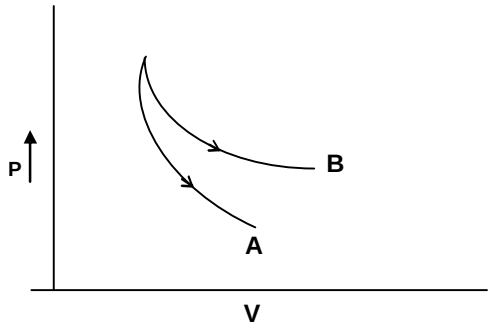


(A) NOT gate (B) OR gate
(C) NOR gate (D) AND gate

15. Velocity of a small ball of mass 0.3 gm and density 8000 kg/m^3 when dropped in a container filled with liquid becomes constant after some time viscous force on ball is given by (density of liquid is 2000 kg/m^3, g = 10 m/s^2)
(A) 2.25×10^{-3} N (B) $25/3 \times 10^{-3}$ N
(C) 50×10^{-3} N (D) 30×10^{-3} N

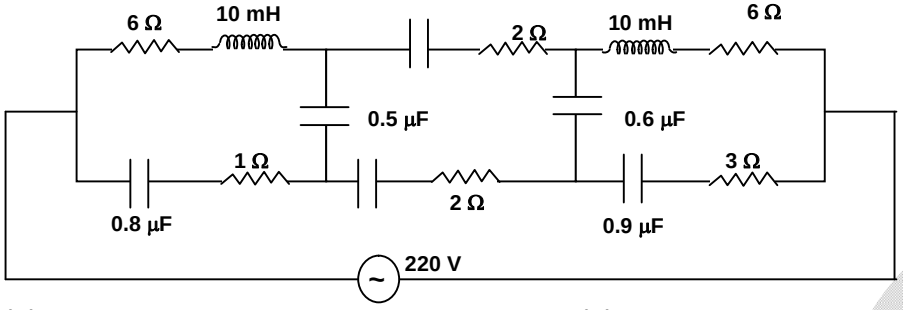
16. Circular rods of different radius r length l and conductivity (k) are present. They are kept at same temperature difference between ends. Which rod will conduct maximum heat.
(A) r = R l = L k = 2k (B) r = 3R, l = 8L k =4k
(C) r = R, l = L/3, k = 2k (D) r = 2R, l = L/2, k = 3k

17. Helium, oxygen and methane gases undergo adiabatic expansions represented by graph. Identify the combination which can **NOT** represent graphs for the gases.



(A) A is for helium, B is for oxygen
(B) A is for oxygen, B is for methane
(C) A is for oxygen, B is for helium
(D) A is for Helium, B is for methane

18. Current in given circuit at very high frequencies



(A) 22 A (B) 44 A
(C) 6A (D) cannot be calculated

19. Two spherical stars A and B emit radiation. The radius of A is 4000 times that of B. A emits 10^6 times the power emitted from B. The ratio of λ_A/λ_B. Where λ_A and λ_B are wavelength of A and B for which radiation power is maximum.
(A) 6 (B) 4
(C) 1 (D) 2

20. A Screw Gauge of pitch 2mm is used to measure the diameter of uniform wire of length 6.8 cm. The main scale reading is in between 1 mm and 2mm and circular scale reading is 67. Find the curved surface area of wire to correct significant figure in cm^2 (Screw gauge has 200 division on circular scale)
(A) 3.0 (B) 4
(C) 3.56 (D) 3.6

SECTION – B

(Numerical Answer Type)

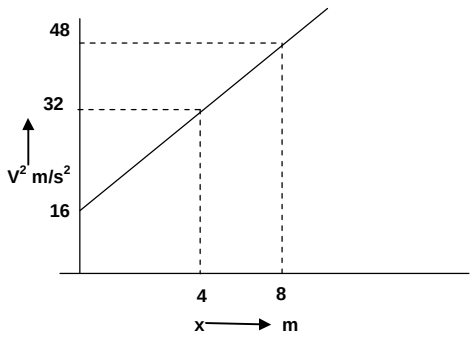
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A string of length L fixed at the two ends is oscillating such that the point at 3L /8 is on antinode. What is least number of loops possible in the string?

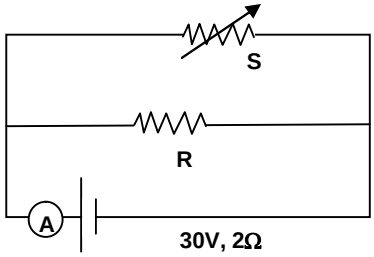
22. Electric lamp of 100 W can be kept glowing by fusion of 2 kg of deuterium. The fusion reaction can be taken as
$${}_1^2H + {}_1^2H \rightarrow {}_1^2He + n + 3.2MeV$$
Time for which electric lamp will glow is
$X \times 10^4$ years. Find out x.

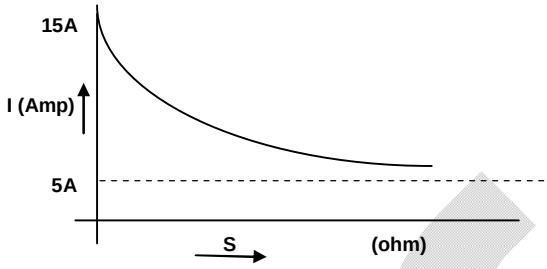
23. Electron in He^+ ion makes a transition from $n = n_1$ excited state to $n = n_2$ excited state. The time period of revolution of electron in initial state is 8 times final state. Minimum value of $n_1 - n_2$ is

24. A particle is moving along x-axis. Following graph shows variation of $(\{\text{velocity } (v)\}^2$ versus displacement(x), then velocity gradient at t = 0.5 sec (if at x = 0, t = 0) is $\frac{N}{10}$ sec^{-1}. Find out N.

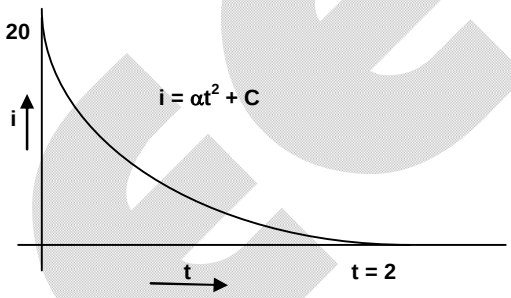


25. An ideal ammeter is connected in series with variable resistance S and resistance R as shown in diagram. Which is connected with all of 30 V and 2Ω internal resistance, ammeter reading is plotted with resistance S. Find out resistance R is ohm.



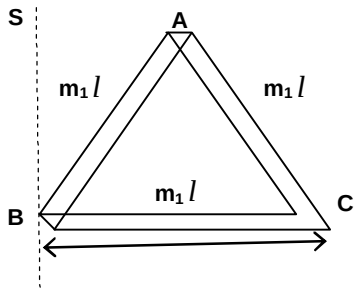


26. Due to change in magnetic flux linked with a coil of resistance 0.3Ω, current is induced in it. The variation of induced current i (in ampere) with time t in (seconds) is shown in diagram. Find out change in magnetic flux, t = 0 to t = 2 sec.



27. 2mW monochromatic light of wavelength 2500 A^0 is incident on photo electric plate. If photoelectric current is measured equal to $I_0\mu A$. The ratio of no of photoelectron emitted and no of photon incident is $P \times 10^{-4}$ then find out P [$I_0 = \frac{1}{3.1}$]

28. Three identical rods are joined in form of equilateral triangle mass and length of each rod is m and length l Moment of inertia about line SB is $\frac{kml^2}{3}$ then find k.



29. Monoatomic gas perform $\frac{Q}{6}$ amount of work. Where Q is heat supplied to gas, then molar heat capacity of gas for the process is given by $\frac{aR}{5}$. Find out a.(R is gas constant)

30. Minimum thickness of mica sheet having $\mu = 1.5$ which should be placed in front of one of the slits in Young double slit experiment is required to reduce the intensity at centre of screen to half of maximum intensity is $n \times 10^{-7}$ m [Assume wavelength is 6000 Å]. Find out n.

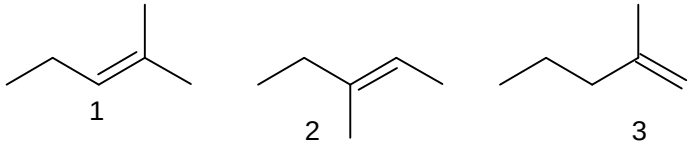
Chemistry

PART – B

SECTION – A
(One Options Correct Type)

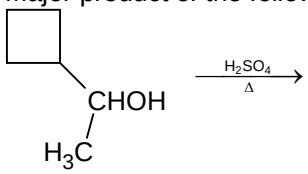
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

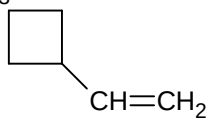
31. Acid-catalysed hydration of an unknown compound X, C_6H_{12}, yields as the major product a racemic mixture Y, $C_6H_{14}O$. Which (if any) of the following is(are) likely for X?

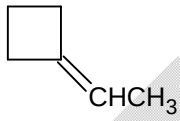


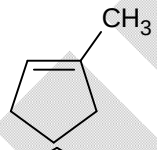
(A) Only 3
(B) Only 2
(C) 1 and 3
(D) None of these

32. Major product of the following reaction is



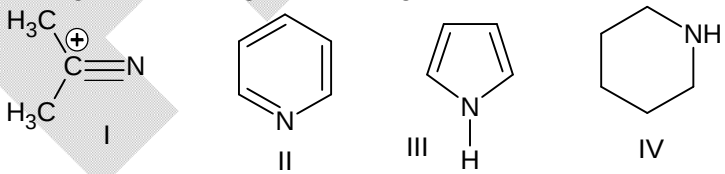
(A)


(B)


(C)


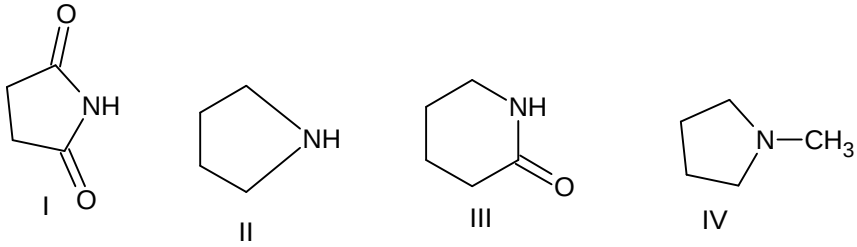
(D)


33. Arrange the following in increasing basic nature



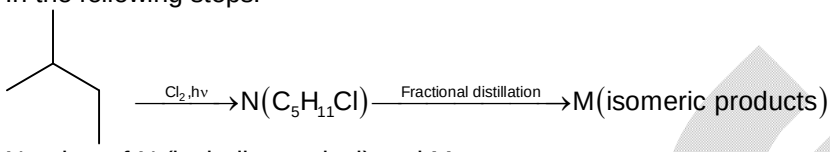
(A) I < II < III < IV
(B) IV < III < II < I
(C) I < III < IV < II
(D) I < III < II < IV

34. Arrange the following in increasing pK_a values



(A) I < II < IV < III
(B) I < III < II < IV
(C) IV < II < III < I
(D) II < III < I < IV

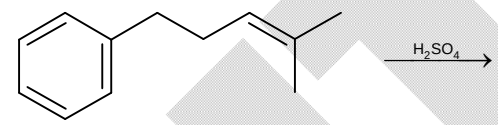
35. In the following steps:



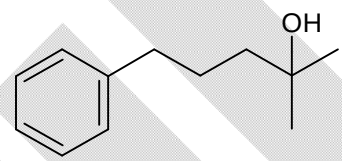
Number of N (including optical) and M are
(A) 6, 6
(B) 6, 4
(C) 4, 4
(D) 3, 3

36. An optically active alkyl bromide $X(C_6H_{13}Br)$ upon treatment with ethanolic KOH solution forms two alkenes Y and Z with their molecular formula (C_6H_{12}). Y and Z are positional isomers. Z upon treatment with cold, dilute alkaline $KMnO_4$ solution gives a *meso*-diol. Hence, X is
(A) 2-bromohexane
(B) 1-bromo-2, 3-dimethyl butane
(C) 3-bromohexane
(D) 2-bromo-2, 3-dimethyl butane

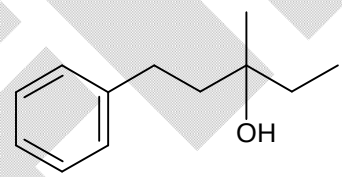
37. End product of the following reaction is



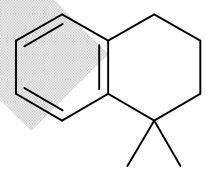
(A)



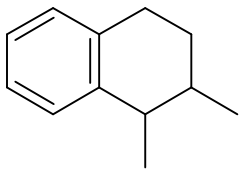
(B)



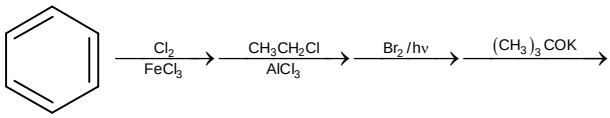
(C)



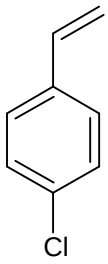
(D)



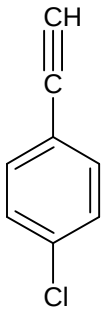
38. End product of the following reaction is



(A)



(B)



(C)



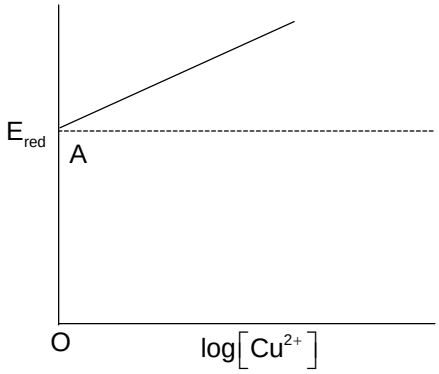
(D)



39. A 0.0020 m aqueous solution of an ionic compound $Co(NH_3)_5(NO_2)Cl$ freezes at $-0.00732^{\circ}C$. Number of moles of ions which 1 mol of ionic compound produces on being dissolved in water will be: $(K_f = +1.86^{\circ}C/m)$

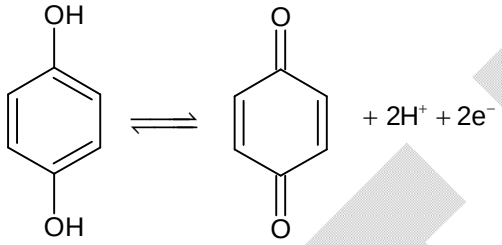
(A) 1
(B) 2
(C) 3
(D) 4

40. $Cu^{2+} + 2e \longrightarrow Cu$; $\log\left[Cu^{2+}\right]$ vs. E_{red} graph is of the type as shown in figure where OA = 0.34 V, then electrode potential of the half cell of $Cu \mid Cu^{2+}(0.1\,M)$ will be:



(A) $-0.34 + \frac{0.0591}{2}$ V
(B) 0.34 + 0.0591 V
(C) 0.34 V
(D) None of these

41. At pH = 2, $E^{o}_{Quinhydrone} = 1.30$ V, $E_{Quinhydrone}$ will be:



(A) 1.36 V
(B) 1.30 V
(C) 1.42 V
(D) 1.20 V

42. Ethylene is produced by cyclobutane is:

$C_4H_8 \xrightarrow{\Delta} 2C_2H_4$

The rate constant is 2.48×10^{-4} sec^{-1}. In what time will the molar ratio of the ethylene to cyclobutane is reaction mixture attain the value 1?

(A) 27.25 minute
(B) 28.25 minute
(C) 25 minute
(D) 20 minute

43. How much faster would a reaction proceed at 25°C than 0°C if the activation energy is 650 kJ?

(A) 2 times
(B) 16 times
(C) 10.4 times
(D) 6 times

44. Which of the following equations is not correctly formulated?

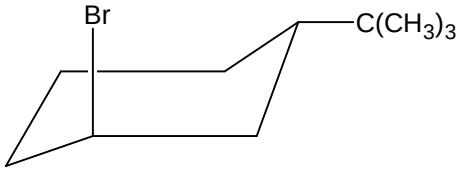
(A) $3Cu + 8HNO_3(dil.) \longrightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$

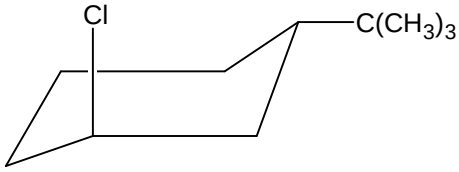
(B) $3Zn + 8HNO_3(very\ dil.) \longrightarrow 3Zn(NO_3)_2 + 2NO + 4H_2O$

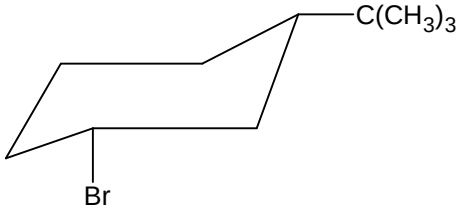
(C) $4Sn + 10HNO_3(dil.) \longrightarrow 4Sn(NO_3)_2 + NH_4NO_3 + 3H_2O$

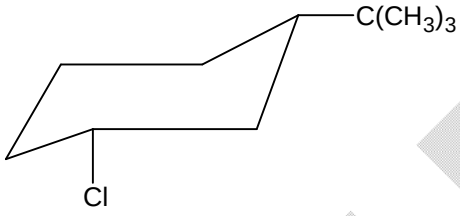
(D) $As + 3HNO_3(dil.) \longrightarrow H_3AsO_3 + 3NO_2$

45. Which alkyl halide undergoes E2 elimination at the fastest rate?

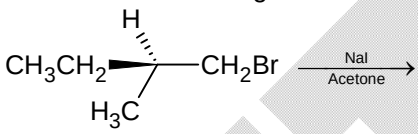
(A) 

(B) 

(C) 

(D) 

46. Consider the following reaction:



Select correct statement:
(A) If follows S_N2 mechanism.
(B) Configuration of the product is the same as the configuration of the reactant.
(C) Both of the above are correct statements.
(D) None of the above is the correct statement.

47. The shortest wavelength transition in the Paschen series in hydrogen occurs at 821 nm; at what wavelength does it occur in Li^{2+} (isoelectronic of hydrogen)?
(A) 91.2 nm (B) 273.6 nm
(C) 821.0 nm (D) 7389.0 nm

48. A gaseous phase reaction taking place in 1 L flask at 400 K is given,
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$.
Starting with 1 mole N_2 and 3 moles of H_2, equilibrium mixture required 250 mL of 1 M H_2SO_4.
Thus, K_c is
(A) 0.006 (B) 0.08
(C) 0.029 (D) 2.05

49. In the following equilibrium amount of CO(g) can be decreased by

$2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$

(A) decreasing volume at constant temperature
(B) adding argon gas at constant pressure
(C) adding O_2 gas
(D) all of the above

50. One mole of an ideal gas is allowed to expand reversibly and adiabatically from temperature 300 K. The work done is 3 kJ. The final temperature of the gas is equal to $\left(C_v = 20\ J\ mol^{-1}\ K^{-1}\right)$

(A) 75 K (B) 150 K
(C) 225 K (D) 290 K

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. The freezing point of an aqueous solution of 0.200 molal KCN solution was -0.744^oC. On adding 0.100 mole $Hg(CN)_2$, the freezing point increased by 0.372^oC. Assume $Hg(CN)_2$ and KCN form complex and KCN is 100% ionized, what is the van't Hoff factor when complex formation takes place?

52. If 3-methyl-2-pentanol is heated with conc. H_2SO_4, then the number of alkenes (including stereoisomers) is……

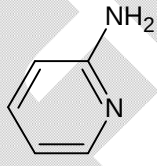
53. Hydrocarbon (A) with two terminal alkynes is subjected to following reaction:

$CH \equiv C(CH_2)_n C \equiv CH \xrightarrow[(II)\ H_2O_2/OH^-]{(I)\ BH_3} (B)$

0.78 g of A gives 1.14 g of B. Thus, molar mass of A is …… g mol^{-1}.

54. (C – Cl) bond length in CH_3Cl is 178 pm. Its dipole moment is 1.87 D. Thus percentage ionic character is …….%

55. 0.94 g of the compound A (given below) was digested with H_2SO_4 and NaOH added to form NH_3, which was absorbed into 0.5 M H_2SO_4 to neutralize the NH_3 completely. Thus, volume of H_2SO_4 (in mL) required to neutralize NH_3 is……..



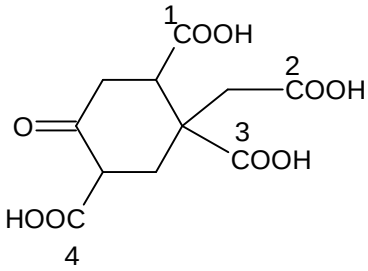
(A)

56. When 200 mL of 0.01 M $Ca(OH)_2$ solution is saturated with 100 mL of CO_2 gas at 29^oC and 0.1 bar. The pH of solution is found to be 1.22×10^x. Find the value of x?

57. Fructose is subjected to bond cleaves by HIO_4. The number of HCHO unit(s) formed per unit of fructose is…….

58. A body fluid is to be analysed for the presence of urea 1 mL of it was decomposed by NaOBr and volume of N_2 gas collected was found to be 5.6×10^{-3} mL at NTP. What is the amount of urea in ppm of body fluid.

59. Which – COOH is lost due to heating?



60. A complex ion $\left[Fe(H_2O)_6\right]^{2+}$ is found to be paramagnetic. It magnetic moment is 4.89 BM. How many unpaired electrons are present?

FIITJEE

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If $x^2 - 2x\cos\theta + 1 = 0$, then the value of $x^{2n} - 2x^n \cos n\theta + 1$, $n \in N$ is equal to
(A) $\cos 2n\theta$ (B) $\sin 2n\theta$
(C) 0 (D) some real number greater than 0

62. Vertices of a parallelogram taken in order are A(2, –1, 4), B(1, 0, –1), C(1, 2, 3) and D. Distance of the point P(8, 2, –12) from the plane of the parallelogram is
(A) $\frac{4\sqrt{6}}{9}$ (B) $\frac{32\sqrt{6}}{9}$
(C) $\frac{16\sqrt{6}}{9}$ (D) none of these

63. If α and β are the real roots of the equation $x^2 - (k - 2)x + (k^2 + 3k + 5) = 0$ ($k \in R$), then the maximum and minimum values of $(\alpha^2 + \beta^2)$ are
(A) 18, $\frac{50}{9}$ (B) 18, $\frac{25}{9}$
(C) 27, $\frac{50}{9}$ (D) none of these

64. The sum of the coefficient of all terms in the expansion of $(2x - y + z)^{20}$ in which y do not appear at all while x appears in even powers and z appears in odd powers is
(A) 0 (B) $\frac{2^{20} - 1}{2}$
(C) 2^{19} (D) $\frac{3^{20} - 1}{2}$

65. All the five digit numbers in which each successive digit exceeds its predecessor are arranged in the increasing order. The $(105)^{th}$ number does not contain the digit
(A) 1 (B) 2
(C) 6 (D) none of these

66. Three people each flip two fair coins. The probability that exactly two of the people flipped one head and one tail, is
(A) $\frac{1}{2}$ (B) $\frac{3}{8}$
(C) $\frac{5}{8}$ (D) $\frac{3}{4}$

67. If $\vec{a},\vec{b},\vec{c}$ are non-coplanar unit vector such that $\vec{a}\times(\vec{b}\times\vec{c})=\frac{1}{\sqrt{2}}(\vec{b}+\vec{c})$ then the angle between the vectors $\vec{a}$ and $\vec{b}$ is

(A) $\frac{3\pi}{4}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{8}$ (D) $\frac{\pi}{2}$

68. $\frac{29\int_0^1(1-x^4)^7\,dx}{4\int_0^1(1-x^4)^6\,dx}=$

(A) 28 (B) 1

(C) 30 (D) 7

69. Let P = (–1, 0), Q = (0, 0) and R = $(3,3\sqrt{3})$ be three points. The equation of the bisector of the angle PQR is

(A) $\frac{\sqrt{3}}{2}x+y=0$ (B) $x+\sqrt{3}y=0$

(C) $\sqrt{3}x+y=0$ (D) $x+\frac{\sqrt{3}}{2}y=0$

70. $\lim_{n\to\infty}\prod_{r=3}^{n}\frac{r^3-8}{r^3+8}$ is

(A) $\frac{2}{7}$ (B) $\frac{7}{2}$

(C) 1 (D) does not exist

71. $\sum_{k=1}^{n}t_k$ is equal to where $t_k=\cot^{-1}\left[\frac{1+k(k+1)^2(k+2)}{2(k+1)}\right]$

(A) $\tan^{-1}((n+1)(n+2))-\tan^{-1}2$ (B) $\tan^{-1}(n+1)-\tan^{-1}2$

(C) $\tan^{-1}2$ (D) $n-\tan^{-1}2$

72. For each positive integer n, let $f(n+1)=n(-1)^{n+1}-2f(n)$ and $f(1)=f(2010)$. Then $\sum_{K=1}^{2009}f(K)$ is equal to

(A) 335 (B) 336

(C) 331 (D) 333

73. A and B are two sets of m distinct elements each. The total number of ways to select subsets A_r and B_r from A and B respectively such that there exists an onto function $f: A_r \to B_r$ is

(A) $\frac{2^{2m}-2^{m+1}+{}^{2m}C_m}{2}$ (B) $\frac{2^{2m}-2^{m+1}+{}^{3m}C_m}{2}$

(C) $\frac{2^{2m}-3^{m+1}+{}^{3m}C_m}{2}$ (D) $\frac{2^{m}-2^{m+1}+{}^{2m}C_m}{2}$

74. $\int \frac{(2+\sqrt{x})dx}{(x+1+\sqrt{x})^2}$ is equal to : (where C is an arbitrary constant)

(A) $\frac{x}{x+\sqrt{x}+1}+C$ (B) $\frac{2x}{x+\sqrt{x}+1}+C$

(C) $\frac{-2x}{x+\sqrt{x}+1}+C$ (D) $\frac{-x}{x+\sqrt{x}+1}+C$

75. The set of all values of k for which the function

$$f(x)=(k^2-3k+2)\left(\cos^2\frac{x}{4}-\sin^2\frac{x}{4}\right)+(k-1)x+\sin 1$$

does not have critical points is

(A) $[1, \infty)$ (B) $(0, 1) \cup (1, 4)$

(C) $(-2, 4)$ (D) $(1, 3) \cup (3, 5)$

76. The locus of the point equidistant from the lines $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ and $\frac{x}{2}=\frac{y}{3}=\frac{z}{1}$ can be

(A) $x + y + 2z = 0$ (B) $x + y - 2z = 0$

(C) $3x + 5y - 4z = 0$ (D) $3x + 5y + 4z + 1 = 0$

77. If y = f(x) is solution of differential equation $\frac{dy}{dx}=\frac{y(1-3x^2y^3)}{x(2x^2y^3+1)}$, then $f^3(x).x^3$ is equal to

(A) $x + cy$ (B) $2x + cy$

(C) $x^2 + cy$ (D) $x^3 + cy^2$

78. $\lim\limits_{x\to 0}\frac{\tan x\sqrt{\tan x}-\sin x\sqrt{\sin x}}{x^3\cdot\sqrt{x}}$ equals

(A) 0.75 (B) 0.85

(C) 0.95 (D) none of these

79. Let $a_n=\int\limits_0^{\pi/2}(1-\sin t)^n\sin 2t\,dt$ then $\lim\limits_{n\to\infty}\sum\limits_{n=1}^{n}\frac{a_n}{n}$ is equal to

(A) 0.15 (B) 0.7

(C) 0.5 (D) 0.12

80. One percent of the population suffers from a certain disease. There is blood test for this disease, and it is 99% accurate, in other words, the probability that it gives the correct answer is 0.99, regardless of whether the person is sick or healthy. A person takes the blood test, and the result says that the he has the disease. The probability that he actually has the disease, is

(A) 0.15 (B) 0.25

(C) 0.56 (D) 0.5

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Given $\vec{A}=2\hat{i}+3\hat{j}+6\hat{k}$, $\vec{B}=i+\hat{j}-2k$ and $\vec{C}=\hat{i}+2\hat{j}+\hat{k}$, then the value of $\left|\vec{A}\times\left[\vec{A}\times(\vec{A}\times\vec{B})\right]\cdot\vec{C}\right|$

82. The mean and standard deviation of 6 observations are 8 and 4 respectively. If each observation is multiplied by 3, then the new standard deviation of the resulting observations is

83. If a, b, c are positive real numbers and $a^2(1 + b^2) + b^2(1 + c^2) + c^2(1 + a^2) = 6abc$, then the value of $\frac{a+b+c}{a^2+b^2+c^2}$ is equal to

84. Let z_1, z_2, z_3 and z_4 be the roots of the equation $z^4 + z^3 + 2 = 0$, the value of $\prod_{r=1}^{4}(2z_r + 1)$ is equal to

85. Let y = f(x) be a real-valued differentiable function on R (the set of all real numbers) such that f(1) = 1. If f(x) satisfies $xf'(x) = x^2 + f(x) - 2$ then find the area enclosed by y = f(x) with x-axis between coordinates x = 0 and x = 3 is

86. Let $A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -x & 14x & 7x \\ 0 & 1 & 0 \\ x & -4x & -2x \end{bmatrix}$ are two matrices such that $AB = (AB)^{-1}$ and $AB \neq I$ (where I is an identity matrix.) Then $Tr(AB + (AB)^2 + (AB)^3 + (AB)^4 + (AB)^5 + (AB)^6)$ is equal to (Tr(A) denotes the trace of matrix A).

87. If A, B and C are angles of a triangle, and $\Delta = \begin{vmatrix} e^{-2i/A} & e^{iC} & e^{iB} \\ e^{iC} & e^{-2iB} & e^{iA} \\ e^{iB} & e^{iA} & e^{-2iC} \end{vmatrix}$. Then $|\Delta|$ is equal to

88. If k is minimum distance of the point (1, 1, 1) from the plane x + y + z = 1 measured perpendicular to the line $\frac{x-x_1}{1} = \frac{y-y_1}{2} = \frac{z-z_1}{3}$. Then [k] is equal to (where [.] is greatest integer function)

89. For any real number α, the parabolas $f_\alpha(x) = 2x^2 + \alpha x + 3\alpha$ passes through the fixed point (a, b). Then the value of $\frac{a+b}{3}$ is

90. Point O is the centre of the ellipse with major axis AB and minor axis CD. Point F is one of the focus of this ellipse. If OF = 6, and the diameter of inscribed circle of ΔOCF is 2, then the value of $\frac{(AB)\cdot(CD)}{13}$ is